



Course Number: PHA G625

Course title: Cellular and Molecular Pharmacology

Instructor In-charge: Dr. Gaikwad Anil Bhanudas

1. Course description: Molecular pharmacology deals with understanding the molecular basis for the actions of drugs and the characteristics of interactions between drug molecules and those of the substrates of drug action in the cell. Course will include chemical, molecular biological and all biochemical and cell biological techniques to understand how cells respond to pharmacologic agents.

2. Scope and objectives of the course:

- To impart advance knowledge about the epigenetic changes involved in the disease development.
- To give advanced knowledge regarding the molecular mechanisms involved in the drug action.

3. Text Book

- Hardman J. G., Limbird L. E. 2011. "Goodman and Gilman's The Pharmacological Basis of Therapeutics" 12th Ed. McGraw-Hill Professional.
- Textbook of Receptor Pharmacology Foreman John C, Johansen Torben, Boca Raton, CRC Press.
- Epigenetics. Allis, Jenuwein, Reinberg and Caparros. Cold Spring Harbour Laboratory Press. ISBN-13: 978-0879697242 Edition: 1

4. Reference Book

- H. P. Rang and M. M. Dale, "Pharmacology" 5th Ed. Churchill Livingstone.
- B. G. Katzung, "Basic and Clinical Pharmacology" 9th Ed. McGraw-Hill Medical.
- Harvey R. A, Champe P. C., "Pharmacology-Lippincott's illustrated Reviews" 4th Ed. Lippincott
- Lodish H, Baltimore D, Berk A, Zipursky SL, Matsundaira P and Darnell J. Molecular Cell Biology. WH Freeman & Company, New York. Latest Edition.
- Brown TA. Gene Cloning and DNA Analysis. Blackwell Science, USA. Latest Edition.
- Alberts B, Bray D, Lewis J, Raff M, Roberts K and Watson JD, (Eds.) Molecular Biology of the Cell. Garland Publishers, New York. Latest Edition.
- Muller RF and Young ID, (Eds.) Emmerly's Elements of Medical Genetics. Churchill Livingstone, New York. Latest Edition.





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5. Additional reading:

- Emerging trends and updates have to be obtained from selected journals, it is mandatory to refer the journals and reviews.
- Students must go through the following journals regularly for current research in areas mentioned in the syllabus.
 - a. Epigenetic
 - b. Nature Medicine
 - c. Nature
 - d. Cell
 - e. Science
 - f. Nature Reviews
 - g. Cancer Letters
 - h. Diabetes
 - i. Diabetes Care

6. Course Plan:

Lectures	Module	Lecture session	Ref	Learning Objective
1-5	Introduction to Molecular Biology and Molecular Genetics	DNA and genome, basic double helix structure, flow of genetic information, molecular basis of transcription and protein synthesis.	Ref book 5 and 6	To learn about the basics of cell and organelles and their functions with respect to protein synthesis and protein transport in a cell
5-15	Endogenous bioactive molecules	Cytokines, neuropeptides and their modulators, neurosteroids, nitric oxide, phosphodiesterase enzyme and protein kinase C, arachidonic acid metabolites, COX-2 regulators and their role in inflammation, endothelium derived vascular substances (NO, endothelins) and their modulators. Pharmacology of atrial peptides, reactive oxygen intermediates, antioxidants and their therapeutic implications.	Text book 1, class notes and research articles	To understand the role of secondary messengers in the in various pathophysiology like diabetes, hypertension atherosclerosis etc.





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15-20	Introduction to Epigenetics	Epigenetic modifications and their function in regulating gene expression, DNA methylation, covalent histone modifications, histone variants, chromatin structure	Text book 3	To understand the role different histone modifications, DNA methylation in transcription.
20-35	Epigenetic deregulation in diseases.	Diabetes: Definition, Genetics and Pathogenesis Oxidative stress in diabetes and its markers, different pathways of oxidative stress in diabetic complications. Cancer Definition, Genetics and Pathogenesis. Metabolic disorders,	Text book 3 and research articles	To study the extraction, separation, transfer, detection of histone using the molecular biology techniques. Roles of these histone modifications in the development of metabolic diseases
35-37	Apoptosis	Pharmacological and clinical implications	Class notes and research papers	Understanding the programmed cell death
37-40	Molecular mechanism of drug action	Receptor occupancy and cellular signaling systems such as G-proteins, cyclic nucleotides, calcium and phosphatidyl inositol. Ionic channels and their modulators.	Text Book 1	Understanding the molecular mechanism with respect to the receptor in drug action. Receptor regulations and phosphorylation.

7. Evaluation Scheme:

No.	Evaluation Component	Duration	Weight-age (%)	Date & Time	Nature of Component
1	Mid Sem Examination	90 min	30		CB
4	Continuous Assessment*		40		OB
5	Comprehensive Exam	120min	30		CB

*Continuous assessment will be based on theory covered in class. Topics and number will be announced in the class. It will be in terms of home assignments, tutorials, projects, laboratory, viva- voce and presentation/ seminars.





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Chamber Consultation Hour: To be announced in the class.

Make-up policy: Generally, make-up will be considered for regular students only (80% attendance in lecture classes). It is solely dependent on the “genuineness” of the circumstances. The make-up application should be personally given to instructor-in-charge.

Notices: Concerning this course will be displayed on Pharmacy Dept. notice board only.

Instructor In charge
PHA G625



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